# BUILD WITH भारत 4.0

NATIONAL LEVEL HACKATHON

**TEAM NAME:** HackX

**PROBLEM STATEMENT:** Smart Campus: AI-Powered Lost & Found Platform with Dynamic Quiz Verification

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**COLLEGE:** Vivekananda Institute of Professional Studies – Technical Campus

# PROBLEM STATEMENT

## Campus Lost & Found is Completely Broken.

Why current informal campus channels fail students at every step:

### THE CYCLE OF FAILURE IN CURRENT CAMPUS LOST & FOUND:

**1. Item Misplaced**
Student leaves earphone/gadget in Library, Canteen, or Lab.

→

**2. Scattered Check**
Student retraces steps; guard cabins have no unified log.

→

**3. WhatsApp Post**
Photo broadcasted on unofficial student WhatsApp groups.

→

**4. Message Flood**
Post gets drowned under 200+ messages within an hour.

→

**5. Privacy Exposure**
Finder/Loser phone number leaked; spam & harassment.

→

**6. False Claim**
Anyone claims valuable items without proof. Never returned.

**CRITICAL BREAKDOWN POINTS:**

- No Central Indexing: Belongings rely on sheer luck to be spotted.
- High Privacy Risk: Students broadcast personal contact info.
- Zero Ownership Verification: First person to send a DM takes the item without verification.

### • Unindexed WhatsApp Chaos

Photos and notices get drowned under hundreds of chat messages within minutes. There is no searchable database or central record.

### • Severe Privacy Leaks

Students publicly post personal phone numbers on broad groups, exposing themselves to unsolicited spam, calls, and safety risks.

### • Zero Ownership Verification

Anyone seeing an item photo can walk up to guard desks and claim it. Without verification, false claims and theft are inevitable.

# SOLUTION

## FindBack: Intelligent Campus Recovery

Multimodal Gemini AI photo inspection combined with dynamic quiz verification.

### ⚡ 3-Second Gemini Vision Inspection

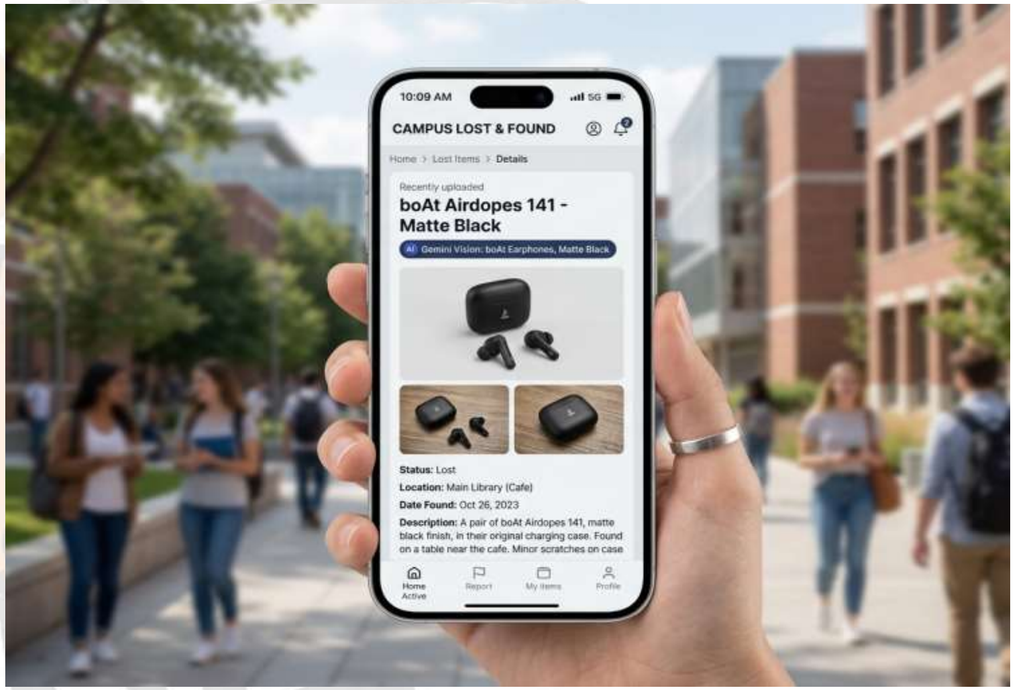
Finder snaps 4-5 photos from different angles. Google Gemini Vision API instantly analyzes the item to extract category, brand, exact color, and condition without tedious manual forms.

### □ AI is the Security Wall (Dynamic Quiz)

Gemini automatically formulates verification questions based on visual features in the photos (e.g. unique stickers, scratches, case color). Only the true owner can answer correctly.

### □ Closed Campus Trust Network (Firebase Auth)

All users must authenticate with verified college student IDs. Eliminates anonymous scammers and ensures complete privacy—students never need to share personal phone numbers.




### Mobile-First PWA: Instant 4-5 Photo Capture

Students upload 4-5 multi-angle photos directly on their phone. Google Gemini Vision auto-tags items and generates verification parameters in seconds.

# FLOW OF SOLUTION

End-to-End System Flow: From 4–5 photo upload to AI quiz verification and safe campus return.

### 1. Snap & Upload (Finder)
- Finder snaps 4–5 multi-angle photos on phone
- Verified College ID login (@vips.edu)
- Zero manual form-filling; instant upload

### 2. Gemini Vision Analysis
- Studies item category, brand, and color
- Extracts hidden scratches, stickers, marks
- Stores visual traits in Firebase Firestore

### 3. Dynamic Quiz Gatekeeper
- Photos stay 100% masked from public
- AI auto-generates 3 visual quiz questions
- Seeker answers based on their true memory

### 4. Verified Campus Return
- Gemini compares response vs photo traits
- Unlocks secure in-app campus coordination
- Zero personal phone numbers shared

## Public Masked Catalog (Zero-Knowledge Listing)

Items listed by generic category only. Photos & identifiers stay hidden to block fake claims:

**🎧 Earphone** | Found near Central Library (2nd Floor)
**Status: Masked Details • [Verify to Claim]**

**📱 Phone** | Found in Student Canteen Seating Area
Status: Masked Details • [Verify to Claim]

**⚽ Sports Ball** | Found near Main Campus Basketball Court
Status: Masked Details • [Verify to Claim]

**💻 Laptop** | Found in Advanced Computer Lab 304
Status: Masked Details • [Verify to Claim]

🔒 Why Blind Catalog? Public photos let opportunistic strangers memorize visual details. Masking forces claimants to answer questions only the true owner knows.

## The AI Security Wall: Verification in Action

Student B clicks 'Claim Earphone' -> Gemini evaluates their response against photo parameters:

**Gemini Visual Ground Truth (Stored Securely • Hidden from Public):**
- Item Inspected: boAt Airdopes 141 (Matte Black)
- Unique Visual Markers: Silver crescent moon sticker on lid + hairline scratch near LED

**Dynamic Quiz Generated by Gemini (Presented to Claimant):**
- **AI Question: 'What unique sticker or physical mark is on the charging case?'**
- Claimant Answer: 'A small silver crescent moon sticker on the top lid.'
- **AI Multimodal Evaluation: Semantic Match Confidence = 98.4% (PASS ✓)**

**✓ VERIFIED: OWNER CONFIRMED**
- Safe in-app handover unlocked.
- Return arranged at Campus Desk.
- Zero phone numbers shared.

**✗ REJECTED: FAILED CLAIM**
- Immediate claim rejection.
- 0 photos or hints revealed.
- Suspicious student ID flagged.

# TECH STACK

## • Technologies to be Used:-

- **Frontend:-** React.js / Next.js 14, Tailwind CSS, Responsive PWA
- **Backend:-** Google Firebase (Cloud Firestore, Cloud Functions)
- **AI Engine:-** Google Gemini 1.5 Flash API (Multimodal Vision & Quiz)
- **Storage:-** Firebase Cloud Storage (Secure 4–5 photo uploads)
- **Security:-** College Institutional Domain Auth, Firestore Security Rules
- **DevOps & CI/CD:-** Firebase Hosting / Vercel Edge, GitHub

| FRONTEND | BACKEND | DATABASE | AI ENGINE | CLOUD & TOOLS |
|---|---|---|---|---|
| React.js | Google Firebase | Cloud Firestore | Gemini API | GitHub |
| Tailwind CSS | Node JS | Firebase Storage | Python SDK | Cloud Functions |
| Next.js 14 | | | | |

## • Core System Modules:-

- **Mobile-First PWA (React & Tailwind):** Clean, accessible mobile interface for photo capture & catalog viewing.
- **Firebase Cloud Backend:** Real-time Firestore database syncing lost/found items and serverless event triggers.
- **Gemini Multimodal AI Engine:** Vision inspection across 4–5 photos and dynamic quiz generation to verify claimants.

## • Process Flow :-

Requirement Gathering → UI Prototyping (Figma) → Firebase & PWA Setup → Gemini Vision Integration → Dynamic Quiz Testing → Campus Rollout

# UNIQUE SELLING PROPOSITION (USP)

## Why FindBack Wins: Real Innovation

Purpose-built technologies that transform lost & found from chaos into trust.

- **The AI Security Wall (Dynamic Quiz)**

  **Solves opportunistic theft and false claims permanently.**

  Google Gemini inspects uploaded photos to dynamically quiz claimants about specific hidden visual markers. True owners get verified instantly; impostors are completely blocked.

- **Multimodal Gemini Vision Auto-Extraction**

  **Eliminates tedious manual reporting forms completely.**

  A quick set of 4-5 photos auto-classifies category, model, and color in seconds via Gemini 1.5 Flash Vision, slashing upload time to under 30 seconds.

- **Verified Student Trust Network (Firebase Auth)**

  **Protects student privacy and builds campus accountability.**

  Both finders and claimants authenticate with verified institutional college IDs. Eliminates public phone number leaks, spam, and unverified strangers.




### AI Verification Quiz: Zero False Claims

Gemini generates dynamic verification questions based on visual features. Only verified owners can answer correctly and initiate item return.

# FEASIBILITY & COMPETITORS

| Key Parameter | WhatsApp Groups | Guard Desks | Notice Boards | FindBack (HackX) |
|---|---|---|---|---|
| **Search & Discovery** | Unindexed Chat Chaos ✗ | Physical Check Only ✗ | Manual Posters ✗ | **Categorized Instant Search ✓** |
| **Ownership Proof** | None (Honor System) ✗ | Subjective Guesswork ✗ | None ✗ | **AI Dynamic Photo Quiz ✓** |
| **Student Privacy** | Phone Number Leaked ✗ | Paper Register ⚠ | Publicly Exposed ✗ | **100% Masked & Protected ✓** |
| **False Claim Defense** | None (First to message) ✗ | Weak / Easily Tricked ✗ | None ✗ | **AI Feature Comparison ✓** |
| **User Accountability** | Anonymous / Open ✗ | Unverified Strangers ✗ | Unverified ✗ | **Verified Campus Student ID ✓** |

## Technical Feasibility

• Cloud-Native Architecture: Built on React/Next.js and Google Firebase. Cloud Firestore handles real-time syncing while serverless functions process requests with zero campus hardware.

• Scalable Multimodal AI: Google Gemini 1.5 Flash provides sub-second visual attribute extraction and dynamic quiz generation via managed API, eliminating complex local model hosting.

## Operational & Campus Adoption

• Free & Frictionless: Accessible to all students directly via mobile web or college portal with zero mandatory app download.

• Accountable Environment: Closed institutional access via verified student IDs prevents non-students and scammers from entering.

# RESEARCH & REFERENCE

## • Campus Observations & Ground Reality:-

• **WhatsApp Clutter:** Unofficial groups fail because photos get buried under unrelated chatter.

• **Privacy Hesitation:** Students, especially women, hesitate to post personal phone numbers publicly.

• **Unclaimed Deadlock:** Items left at guard desks remain uncataloged and are rarely reunited with owners.

## • Technical & Academic Foundations:-

• **Google Gemini Multimodal Vision:** Enables instant item categorization and visual attribute correlation across 4-5 photos.

• **Real-Time Document Indexing (Firestore):** Combines instant status synchronisation with structured campus metadata.

• **Digital Campus Directives:** Aligned with modern student welfare and smart campus governance directives.

## • Phased Implementation Roadmap:-

• **Phase 1: Working Prototype (Current):** Core photo upload, AI visual inspection, and dynamic quiz verification.

• **Phase 2: College Campus Pilot:** Rollout at college campus with student ID authentication and library desks.

• **Phase 3: Centralized Security Desks:** Providing an intuitive catalog dashboard for campus guards and admin.

• **Phase 4: Multi-Campus Expansion:** Extending to inter-college fests, sports meets, and university events.

## • Project Summary & Team HackX:-

• **Project Name:** FindBack – Smart Campus Lost & Found Platform

• **Institution:** Vivekananda Institute of Professional Studies – Technical Campus

• **Core Mission:** Eliminating campus chat chaos through AI-driven, privacy-first recovery.

## • Execution Flow :-

**Problem Identification** → **Campus Survey** → **Gap Analysis** → **AI Architecture** → **Prototype Testing** → **Campus Pilot**